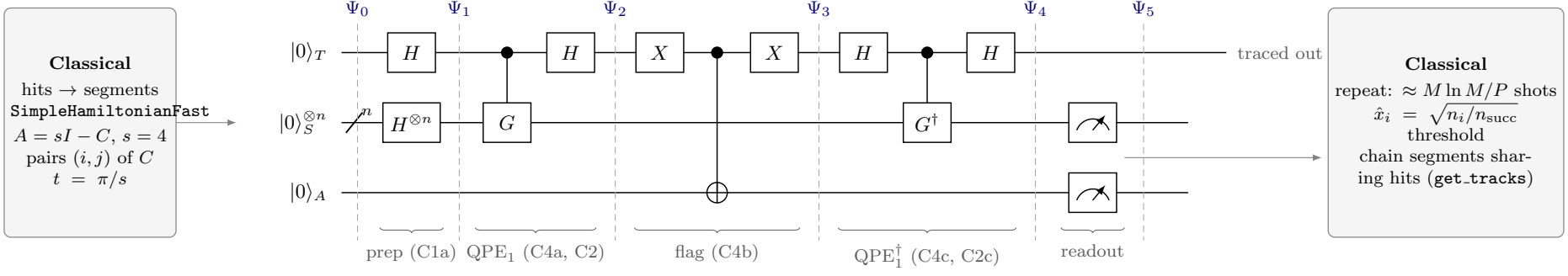
Full computation 1: segment register, one-bit filter (1BQF), as built in `one_bqf.py` (centralise-2.1)



$$\Psi_1 = |0\rangle_A|+\rangle_T|u\rangle_S,\ \ |u\rangle = \textstyle\sum_j \beta_j|u_j\rangle \qquad \Psi_2 = |0\rangle_A \textstyle\sum_j \beta_j e^{-i\lambda_j t/2}\big(c_j|0\rangle + is_j|1\rangle\big)_T|u_j\rangle \qquad \Psi_3 = \textstyle\sum_j \beta_j e^{-i\lambda_j t/2}\big(c_j|1\rangle_A|0\rangle_T + is_j|0\rangle_A|1\rangle_T\big)|u_j\rangle$$

$$\Psi_4 = \textstyle\sum_j \beta_j\big\{|1\rangle_A(c_j^2|0\rangle - ic_js_j|1\rangle)_T + |0\rangle_A(s_j^2|0\rangle + is_jc_j|1\rangle)_T\big\}|u_j\rangle \qquad P = \textstyle\sum_j \beta_j^2 c_j^2,\ \ p_i = \big[\cos^2(\tfrac{At}{2})u\big]_i^2 + \big[\tfrac{1}{2}\sin(At)u\big]_i^2 \qquad (c_j = \cos\tfrac{\lambda_j t}{2},\ s_j = \sin\tfrac{\lambda_j t}{2};\ 7.9)$$

Worked event (7.8): $n = 5$, 7 qubits, 6 pairs, 507 CX after transpiling; $P = 0.0842$, all of it on the 8 true segments. Width $n + 2$ for any event; 2 calls of c-G per shot; with amplitude amplification about $\pi/(4\sqrt{P})$ rounds. The studies evaluate p_i directly (7.5) instead of sampling.